Paper 28 - N-Dimensional Game Lattices

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Construction Basis

Extend chess/board logic into arbitrary dimensional local-rule games and tool operators.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-28.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-28\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-28\verify_nd_game_lattices.py
- receipt: production\formal-papers\CQE-paper-28\nd_game_lattices_receipt.json (CQE-paper-28: pass_with_open_obligations)
- body: production\papers\CQE-paper-28\PAPER-BODY.md
- source: production\papers\CQE-paper-28\SOURCE.md
- formal: production\papers\CQE-paper-28\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-28\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-28\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-28.md

Paper 28 - N-Dimensional Game Lattices

Abstract

Paper 28 defines a local-rule game lattice as a finite chart operator. A game piece reads a local `(L, C, R)` neighborhood, emits a Rule 30-style occupancy bit, moves through an S3 trace-2 orbit, logs forbidden carriers as residue, and anneals to a Lie-conjugate state in at most three steps.

The closed result is constructive and bounded. The verifier proves the forced code-tower dimensions `{1,3,7,8,24,72}`, checks the dimension-8 extended Hamming board, enumerates a six-row S3 move orbit for a robot on the dimension 8 board, logs one forbidden straight carrier, and confirms that all orbit rows close under centroid annealing. The paper does not claim to solve chess, Go, arbitrary games, arbitrary dimensions, or any complete game-theoretic problem.

Closed Evidence

The receipt is generated by `production/formal-papers/CQE-paper-28/verify_nd_game_lattices.py`. It imports the live `lattice_codes`, `centroid_voa`, `f4_action`, and `rule30` package surfaces.

`verify_lattice_code_chain` passes. Its forced dimensions are `1`, `3`, `7`, `8`, `24`, and `72`; the powered shortcut is `1`, `9`, `49`, and `72`; and the sheet-K bound is `9`.

`verify_extended_hamming_8` passes with 16 codewords, minimum weight 4, and weight distribution `{0:1, 4:14, 8:1}`. This supplies the worked board's dimension-8 lattice surface.

The finite robot example starts at chart state `(1,0,1)`. The S3 move orbit has six rows and three unique target states: `(0,1,1)`, `(1,0,1)`, and `(1,1,0)`. Five rows are legal orbit moves and one identity carrier is logged as forbidden. Every row closes to a Lie-conjugate state in at most three anneal steps.

Definitions

A local-rule game is a finite receipt `(lattice, neighborhood, move rule, obligation ledger)`. The move rule reads a local `(L, C, R)` chart state and emits an occupancy bit.

An admissible dimension is one of the verified code-tower dimensions `{1,3,7,8,24,72}`. A game on another dimension may exist as a candidate, but it does not inherit this proof surface without its own verifier.

A move orbit is the set of trace-2 states produced by the six S3 permutations. Repeated target states are retained in the receipt because they came from different group elements.

A forbidden carrier is a move row that the game policy excludes. It is logged as a constraint, not deleted.

A closed game solver claim is a claim about strategy, termination, winning states, fairness, or complete game solution. This paper does not make such a claim.

Claims

- The verified code-tower dimensions define admissible game-lattice surfaces.
- Dimension 8 is a valid worked board through the extended Hamming verifier.
- A trace-2 S3 orbit supplies a finite move surface for a local-rule piece.

- Rule 30 local emission gives each orbit row a replayable occupancy bit.
- Forbidden carriers can be logged without deleting the move receipt.
- Every chart row in the receipt closes to a Lie-conjugate attractor in at most three steps.
- General game solving and real-game strategy are open obligations.

Theorem 28

An N-dimensional game lattice is valid in the CQE kernel when it is presented as a finite local-rule receipt on an admissible code-tower dimension: the move orbit is enumerated, emissions are replayable, forbidden carriers are logged, and every row carries its closure or obligation status.

Proof

The dimension claim follows from `verify_lattice_code_chain`. The verifier passes every layer of the chain and returns the forced dimension set `{1,3,7,8,24,72}`. This proves that the game board dimensions used here are not arbitrary choices inside this kernel.

The worked-board claim follows from `verify_extended_hamming_8`. The dimension-8 board has the expected extended Hamming parameters: 16 codewords, minimum weight 4, and weight distribution `{0:1, 4:14, 8:1}`.

The move-orbit claim follows from `S3_PERMUTATIONS` and the trace-2 state map. Starting from `(1,0,1)`, the six S3 elements produce six receipt rows and three unique target states. The identity row is marked `forbidden_logged`; it remains in the receipt as a constraint. The other five rows are legal orbit moves.

The emission claim follows from `rule30_bit` applied to every target state. The closure claim follows from `anneal_to_lie_conjugate`: the maximum anneal count in the worked receipt is three, and the global centroid closure verifier also passes over all eight chart states. Therefore the local-rule game lattice is closed as a finite receipt.

Nothing in the receipt evaluates strategy, game termination, winning states, or arbitrary real-piece geometry. Those are therefore obligations, not hidden claims.

Worked Example

The worked example is a dimension-8 robot with forbidden straight carriers. The active state is `(1,0,1)`. Its legal surface is the S3 orbit of the trace-2 state. The identity carrier is disallowed by policy and logged. The remaining five orbit rows are available moves. Each row records the permutation, target state, Rule 30 emitted bit, anneal count, final attractor, and carrier status.

This is enough to prove a local-rule game-lattice receipt. It is not enough to solve a game.

Open Obligations

The general N-dimensional game solver is not claimed. The receipt proves finite orbit closure, not arbitrary solvability.

Non-code-tower dimensions remain open. Dimension 5 is explicitly rejected by the verifier as outside the inherited proof surface.

Real game-piece geometry remains open. Each piece type needs its own map into the trace-2/S3 orbit.

Complete game theory remains open. Legal move receipts do not prove strategy, termination, winning states, or fairness.

Suite Role

Paper 28 extends Paper 24's board automata into a reusable local-rule game receipt. It gives the suite a controlled way to talk about game-like systems, CAD constraints, robot movement, and search surfaces without pretending that a finite move receipt is a complete solver.